# Final project NLP

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The source code of the project can be viewed on github: https://github.com/GodofRussia/-NLP-Final-Project

## I. INTRODUCTION

This work aims to research IO, BIO and BIOES labeling schemes for transformer-based named entity recognition. In this work we will look at different types of tagging, compare their results using metrics and see which ones are better.

## II. DATASET

The *conll2003 dataset* [1] was selected as main dataset for this work. This dataset contains more than 14,000 samples with ner tags.

The dataset annotates four types of named entities:

- PER (Person): Names of people (e.g., *John Smith*, *Nadim Ladki*).
- ORG (Organization): Companies, government bodies, sports teams, and other institutions (e.g., *United Nations*, *Apple*).
- LOC (Location): Geographical entities such as cities, countries, and natural landmarks (e.g., *Brussels*, *Syria*).
- MISC (Miscellaneous): Entities that do not belong to the previous three categories, including nationalities, events and titles (e.g., *Asian Cup*, *Japanese*).

## III. NER

### A. IO

First possible tagging algorithm is IO. This is one of the simplest and most basic algorithms, which only involves dividing into basic parts. More details can be found in the image 1.

It consists of two tags

- I-(Inside): A token that belongs to an entity (for example, I-PER for any token of a person name);
- O (Outside): A token that does not refer to any entity.

| Words | IO Label | BIO Label | BIOES Label |
|---|---|---|---|
| Jane | I-PER | B-PER | B-PER |
| Villanueva | I-PER | I-PER | E-PER |
| of | O | O | O |
| United | I-ORG | B-ORG | B-ORG |
| Airlines | I-ORG | I-ORG | I-ORG |
| Holding | I-ORG | I-ORG | E-ORG |
| discussed | O | O | O |
| the | O | O | O |
| Chicago | I-LOC | B-LOC | S-LOC |
| route | O | O | O |
| . | O | O | O |

FIG. 1: NER Tagging

In the picture you can see an example 2.

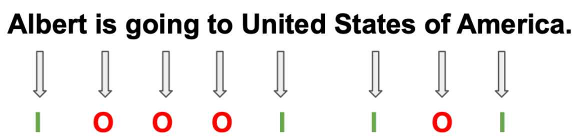


FIG. 2: IO tagging on example

The main weakness of the IO scheme is its inability to distinguish between two adjacent entities of the same type. For example, in the phrase *"John Smith Jane Doe met"* would be four I-Person tokens but we have two differents persons on fact.

### B. BIO

The task of BIO classification is to determine the meaning of a word in a sentence like in IO but in extend form.

In addition to IO tags there is another one - B tag (means Begin), we use it to understand if entity started. All next words in entity will be mentioned by I (Inside) tag.

- B-(Begin): The beginning of an entity (for example, B-PER for the beginning of a name);
- I-(Inside): The internal part of an entity (for example, I-PER for the continuation of a name);
- O (Outside): A token that does not refer to any entity.

Here is an example of sentence tags 3.

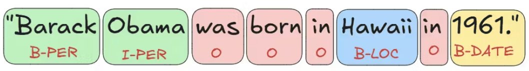


FIG. 3: BIO tagging on example

### C. BIOES

BIOES uses a wide variety of different labels. In addition to BIO, it uses S (Single) to identify a single entity. It also has a different meaning for the tokens B (Begin) - the beginning of a multi-word entity and E (End) - the end of a multi-word entity. All tokens within this entity are marked as I (Inside).

- B-(Begin): The beginning of a multi-word entity (for example, B-PER for the first token of a name);
- I-(Inside): The internal part of a multi-word entity (for example, I-PER for a middle token of a name);
- O (Outside): A token that does not refer to any entity;
- E-(End): The last token of a multi-word entity (for example, E-PER for the last token of a name);
- S-(Single): A single-token entity (for example, S-LOC for a one-word location).

This way the model receives a stronger signal about entity boundaries: every span has both an explicit start and an explicit end and single-word entities are clearly separated from the first token of longer ones.

In the picture you can see an example 4.

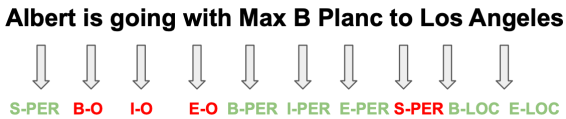


FIG. 4: BIOES tagging on example

## IV. TOKENIZATION

For the tokenization task we will also use `bert-base-cased`.

Tokenization causes problems due to the fact that len(tokens) $>=$ len(words), and we only have markup for words. The problem is illustrated in more detail in the image 5.

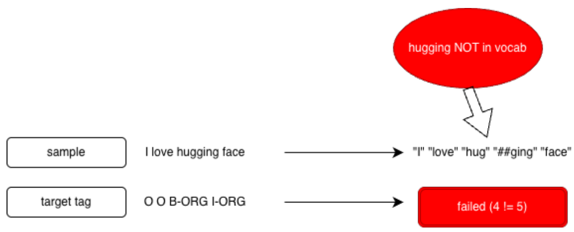


FIG. 5: BIO

To solve this problem, we can either ignore subsequent parts or count them as part of the word (continuation). In our work we will take them into account, described in detail in the image

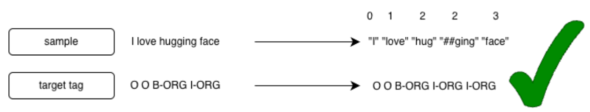


FIG. 6: BIO

## V. TRANSFORMER

In order to use tagging, we will use the BERT encoder-decoder transformer model. To do this, we will use a ready-made BERT model from the transformer library. We will use `AutoModelForTokenClassification` due to our task is classificate every token, not all sentence (otherwise we need `AutoModelForSequenceClassification`)

Listing 1: Importing tokenizer and token classification model from Transformers

```
from transformers import
    AutoTokenizer,
    AutoModelForTokenClassification
```

We will use `bert-base-cased` as main transformer model. The key difference with uncased is that our model will be all case sensitive (upper and lower case letters). This is very useful for our task, because words can have different meanings in different registers.

## VI. TRAINING

To further understand the learning dynamics of each scheme, we analyzed the training loss over time 7.

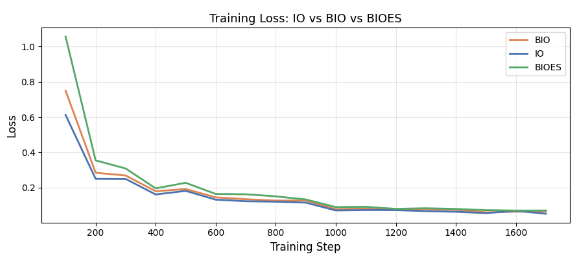


FIG. 7: Training loss comparison across IO, BIO, and BIOES schemes.

The loss curves show that all three models train successfully. However, the BIOES scheme has a slightly higher loss than BIO, and IO has the lowest loss. This is logical: BIOES has more tags (17) compared to BIO (9) and IO (5), making the token classification task harder for the model during training.

## VII. METRICS

To assess the quality of the model, we will use the seqeval library [2], designed for sequence tagging tasks such as named entity recognition. It allows you to calculate precision, recall, F1-score and accuracy metrics based on a comparison of true and predicted token labels. In this work, seqeval is used to analyze how correctly the model determines the boundaries of named entities and assigns them the appropriate classes.

In our measurements, we compared in two approaches. First is overall metrics such as accuracy, precision, F1 for each of the tagging methods (see 8). Second one is F1 score for all 4 entities 9.

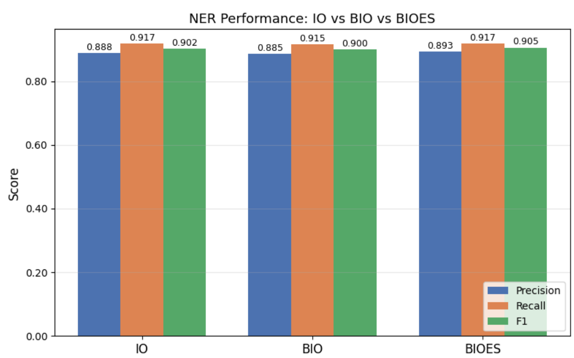


FIG. 8: Overall metric results

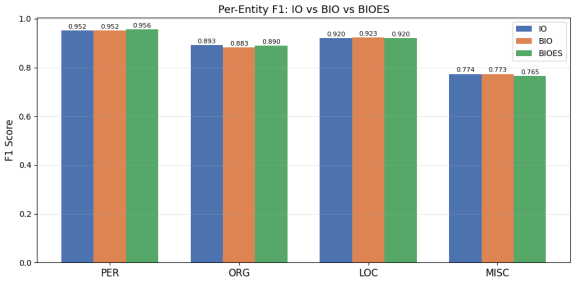


FIG. 9: Per-entity metric results

## VIII. RESULTS

Table 8 presents the overall span-level evaluation metrics computed via micro-averaging across all classes. All three schemes achieve comparable performance, with BIOES yielding a slightly higher overall F1 score and precision.

However, Table 9 reveals the paradox dynamic opposite to the previous graphic when examining per-entity F1 scores. The simpler IO scheme actually outperforms both BIO and BIOES in half of the four entity types: ORG and MISC.

After some thought, we concluded that a fairly logical and consistent picture had emerged. IO wins because the dataset has the simplest markup and the task of capturing the desired sequence using spans. A loss in the PER test could indicate a multi-person merging error in this dataset. However, IO outperforms entities, again due to spans logic of metrics and other entities usual appearances. In the overall metrics, there are errors due to multi-word sequences of different entities, and it's clear that it lags behind BIOES individually. Ultimately, the overall PER percent in the dataset is higher, and the more complex variants win.

BIO wins in LOC, it is important that countries are medium-length entities, so BIO is excellent balanced in computational cost and train complexity.

Table I demonstrates the inference behavior on a sample sentence. All three schemes correctly identify adjacent entities of the same type when separated by punctuation.

TABLE I: Per-token predictions on "AL-AIN , United Arab Emirates".

| Word | Gold | IO | BIO | BIOES |
|---|---|---|---|---|
| AL-AIN | B-LOC | I-LOC | B-LOC | S-LOC |
| , | O | O | O | O |
| United | B-LOC | I-LOC | B-LOC | B-LOC |
| Arab | I-LOC | I-LOC | I-LOC | I-LOC |
| Emirates | I-LOC | I-LOC | I-LOC | E-LOC |

To demonstrate the models' capabilities on unseen data, we tested them on several random sentences using a simple aggregation strategy. Table II shows the extracted entities. All three schemes successfully identify and group multi-word entities and correctly separate adjacent entities of different types.

TABLE II: Inference comparison on random texts across IO, BIO, and BIOES models.

| Text | IO Entities | BIO Entities |
|---|---|---|
| Elon Musk and Jeff Bezos both attended the SpaceX launch in Florida. | Elon Musk [PER]<br>Jeff Bezos [PER]<br>SpaceX [ORG]<br>Florida [LOC] | Elon Musk [PER]<br>Jeff Bezos [PER]<br>SpaceX [ORG]<br>Florida [LOC] |
| The European Union and United Nations held talks in Geneva. | European Union [ORG]<br>United Nations [ORG]<br>Geneva [LOC] | European Union [ORG]<br>United Nations [ORG]<br>Geneva [LOC] |

In III and IV you can see final win distribution between schemes and accuracy that has the highest value in IO.

TABLE III: Overall span-level evaluation results on the CoNLL-2003 test set.

| Scheme | Precision | Recall | F1 | Acc. |
|---|---|---|---|---|
| IO | 0.8879 | **0.9174** | 0.9024 | **0.9722** |
| BIO | 0.8850 | 0.9152 | 0.8998 | 0.9719 |
| BIOES | **0.8931** | **0.9174** | **0.9051** | 0.9705 |

TABLE IV: Per-entity F1 scores on the CoNLL-2003 test set across the three labeling schemes.

| Entity | IO | BIO | BIOES |
|---|---|---|---|
| PER | 0.9515 | 0.9515 | **0.9558** |
| ORG | **0.8926** | 0.8825 | 0.8897 |
| LOC | 0.9202 | **0.9230** | 0.9200 |
| MISC | **0.7736** | 0.7734 | 0.7653 |

We also evaluated the token-level entity-type confusion matrices for all three schemes 10.

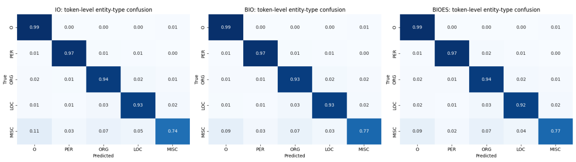

FIG. 10: Token-level entity-type confusion matrices for IO, BIO, and BIOES.

The matrices show that the main problem for all schemes is confusing the `O` (Outside) tag with the `MISC` category. This happens because `MISC` contains very different and complex words. Also, we can see that IO, BIO, and BIOES make similar mistakes when predicting entity types (for example, predicting `ORG` instead of `LOC`). This proves that the tagging scheme mostly helps with boundaries, not with understanding the entity type itself.

Finally, to assess how well the model handles the expanded 17-tag inventory of the BIOES scheme, we plotted the full tag-level confusion matrix 11.

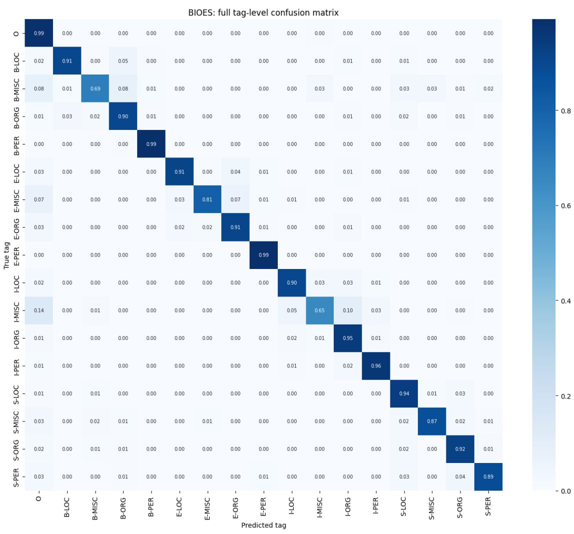

FIG. 11: Full tag-level confusion matrix for the BIOES scheme.

The matrix has a strong diagonal line, which means the model learned all 17 tags very well. The model almost never confuses tags inside the same entity (for example, it doesn't mix `B-PER` with `I-PER`). The most common boundary mistakes happen in the `MISC` class (like predicting `I-MISC` instead of `E-MISC`). This shows again that `MISC` is the hardest category to find exact boundaries for.

## IX. CONCLUSION

The results of our empirical comparison show that IO, BIO, and BIOES labeling schemes achieve nearly identical overall performance when fine-tuning model on the CoNLL-2003 dataset.

Contrary to the common assumption that more expressive tagsets strictly improve NER performance, the simplest scheme, IO, proved to be highly competitive. It achieved the highest overall accuracy and outperformed BIO and BIOES on specific entity types like ORG and MISC (LOC in IO is equal to top-2). This suggests that modern transformer models can largely infer entity boundaries from contextual embeddings alone, without relying heavily on explicit `B-` or `E-` markers.

Our analysis of confusion matrices also supports this. Token-level matrices showed that all schemes make similar mistakes when guessing the entity type (like confusing ORG with LOC), meaning the tagset mostly helps with boundaries, not semantics. Furthermore, the full BIOES matrix proved that the model easily learns a complex 17-tag system without mixing up internal tags (like B-PER and I-PER). The main difficulty for all models remains the MISC category, which causes the most boundary and classification errors.

In addition to empirical metrics, you can also look at the objective part of the problem. IO gives a simple classification - whether a word refers to any entity. For example, we want to get all the medical terms that were mentioned in the text - then we just look at I-TERM.

On the other hand, our task might be to extract the full names of drugs or organizations (multiwords). For these tasks you can use BIO or BIOES. BIOES is better at defining boundaries, but more complex (contains more tags). More complex algorithms are preferable when exact boundary recovery of complex, multi-word entities, despite the cost of doubling the tag inventory.

[1] Huggingface, *conll2003*, https://huggingface.co/datasets/tomaarsen/conll2003 (accessed: 25 April 2026).

[2] Seqeval library, *seqeval*, https://pypi.org/project/seqeval/0.0.10 (accessed: 27 April 2026).